OS-4

U13157005 吳翊弘 子二甲

File Edit Selection View Go Run ... MultiBlink

EXPLORER

MULTIBLINK
- .vscode
- build
- port
- src
  - Agent.cpp 6
  - Agent.h 3
  - BlinkAgent.cpp 3
  - BlinkAgent.h
  - CMakeLists.txt
  - main.cpp 9+
- .gitignore
- CMakeLists.txt
- FreeRTOS_Kernel_import.cmake
- pico_sdk_import.cmake

Agent.cpp 6 | BlinkAgent.cpp 3 | main.cpp 9+ | Agent.h 3

src > main.cpp > ...

```cpp
//U13157005 吴翊弘
#include "pico/stdlib.h"

#include "FreeRTOS.h"
#include "task.h"
#include <stdio.h>

#include "BlinkAgent.h"


//Standard Task priority
#define TASK_PRIORITY       ( tskIDLE_PRIORITY + 1UL )

//LED PAD to use
#define LED_PAD 25
#define LED1_PAD 10
#define LED2_PAD 11
#define LED3_PAD 12
#define LED4_PAD 13
#define LED5_PAD 14
#define LED6_PAD 15
#define LED7_PAD 16
#define LED8_PAD 17



void runTimeStats(   ){
    TaskStatus_t *pxTaskStatusArray;
    volatile UBaseType_t uxArraySize, x;
    unsigned long ulTotalRunTime;


    // Get number of takss
    uxArraySize = uxTaskGetNumberOfTasks();
```

PROBLEMS 28 | OUTPUT | DEBUG CONSOLE | TERMINAL | Compile Project - Task

```
ninja: Entering directory `C:\Users\LAB216\Downloads\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSC
ultiBlink/build'
[2/2] Linking CXX executable src\Blink.elf
* Terminal will be reused by tasks, press any key to close it.
```

OUTLINE
TIMELINE

File Edit Selection View Go Run ... MultiBlink

EXPLORER

MULTIBLINK
- .vscode
- build
- port
- src
  - Agent.cpp 6
  - Agent.h 3
  - BlinkAgent.cpp 3
  - BlinkAgent.h
  - CMakeLists.txt
  - main.cpp 9+
- .gitignore
- CMakeLists.txt
- FreeRTOS_Kernel_import.cmake
- pico_sdk_import.cmake

Agent.cpp 6 | BlinkAgent.cpp 3 | main.cpp 9+ | Agent.h 3

src > main.cpp > ...

```cpp
/***
 * Main task to blink external LED
 * @param params - unused
 */
void mainTask(void *params){
    BlinkAgent blink(LED_PAD);
    BlinkAgent worker1(LED1_PAD);
    BlinkAgent worker2(LED2_PAD);
    BlinkAgent worker3(LED3_PAD);
    BlinkAgent worker4(LED4_PAD);
    BlinkAgent worker5(LED5_PAD);
    BlinkAgent worker6(LED6_PAD);
    BlinkAgent worker7(LED7_PAD);
    BlinkAgent worker8(LED8_PAD);

    printf("Main task started\n");

    blink.start("Blink", TASK_PRIORITY);
    worker1.start("Worker 1", TASK_PRIORITY);
    worker2.start("Worker 2", TASK_PRIORITY + 1);
    worker3.start("Worker 3", TASK_PRIORITY + 2);
    worker4.start("Worker 4", TASK_PRIORITY + 3);
    worker5.start("Worker 5", TASK_PRIORITY + 4);
    worker6.start("Worker 6", TASK_PRIORITY + 5);
    worker7.start("Worker 7", TASK_PRIORITY + 6);
    worker8.start("Worker 8", TASK_PRIORITY + 7);

    while (true) { // Loop forever
        runTimeStats();
        vTaskDelay(pdMS_TO_TICKS(3000));
    }
}
```

PROBLEMS 28 | OUTPUT | DEBUG CONSOLE | TERMINAL | Compile Project - Task

```
ninja: Entering directory `C:\Users\LAB216\Downloads\RPIPicoFreeRTOSCourse8\RPIPicoFreeRTOSCourse8\4-Tasks\M
ultiBlink/build'
[2/2] Linking CXX executable src\Blink.elf
* Terminal will be reused by tasks, press any key to close it.
```

OUTLINE
TIMELINE

CHAT

Welcome to Copilot

Let's get started

Add context (#), extensions (@), commands

Build Workspace | Show Config

Review AI output carefully before use.

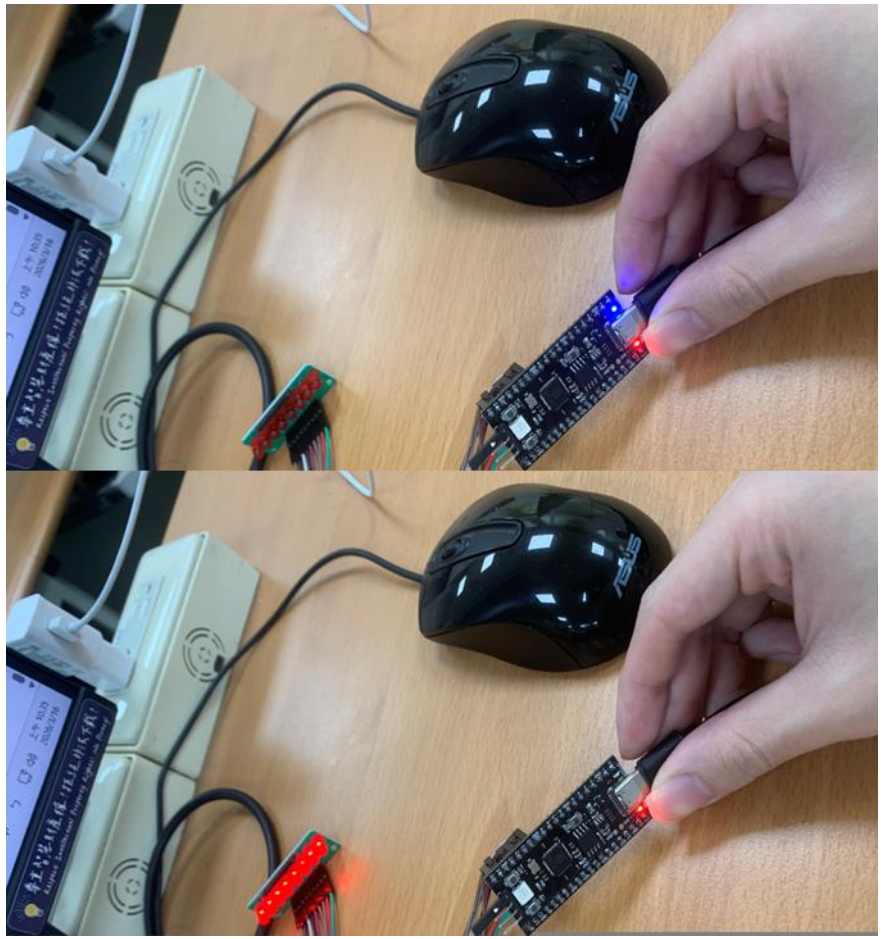
⊗ 28 ⚠ 0 | Ln 152, Col 1 | Tab Size: 4 | UTF-8 | LF | {} C++ | Compile | Run | Pico SDK: 2.2.0 | Board: pico | Pico

RST ànniǔ yīcì hòu, jiāng UF2

READ ME

MultiBlink

- 控制板載 LED (GPIO25) 與 8 顆外接 LED交替閃爍
- 板載 LED 先亮，再依序亮起外接 LED

使用

1. 編譯專案生成 .uf2
2. 按住 BOOT 上傳到 Pico
3. 觀察 LED 閃爍